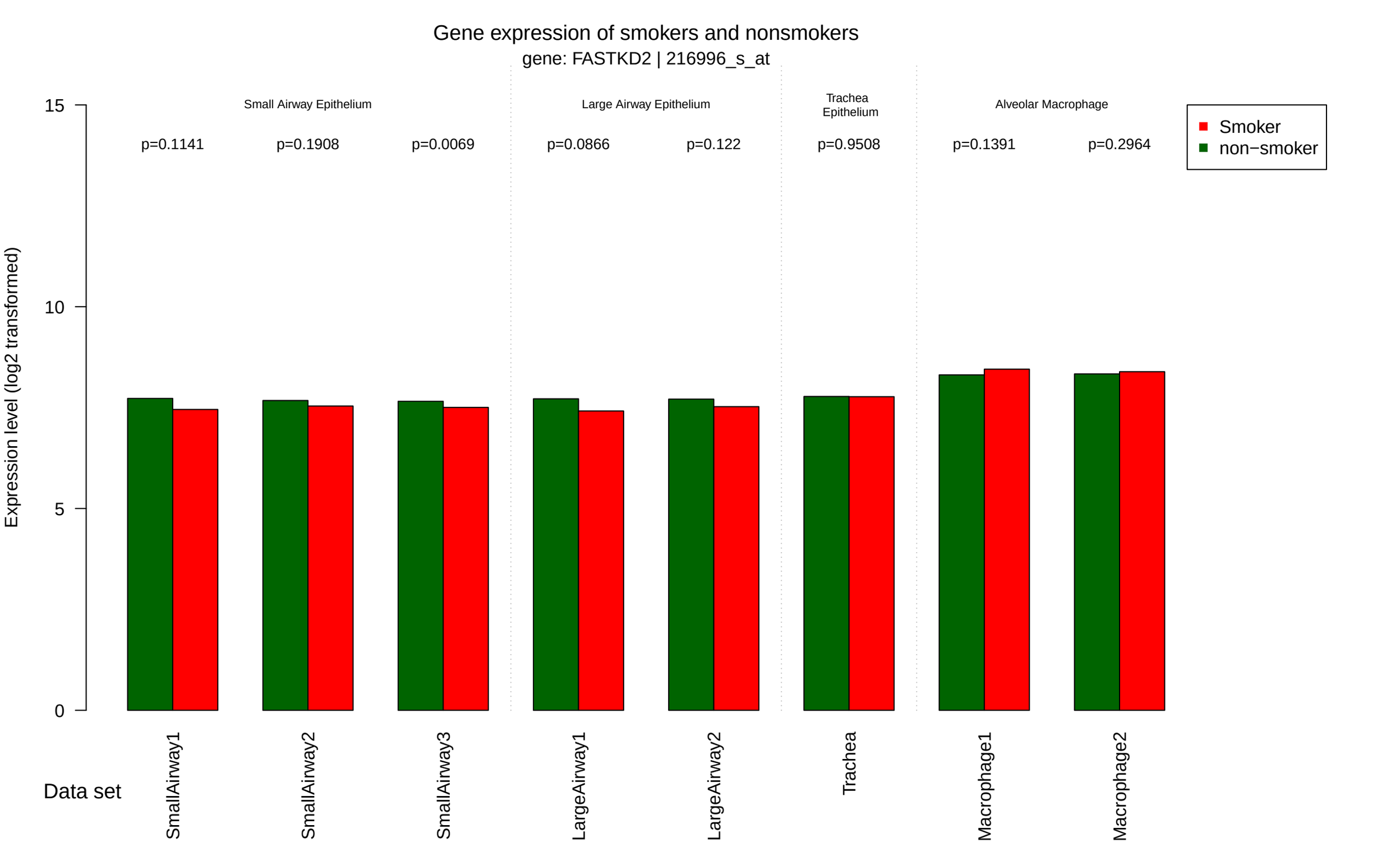
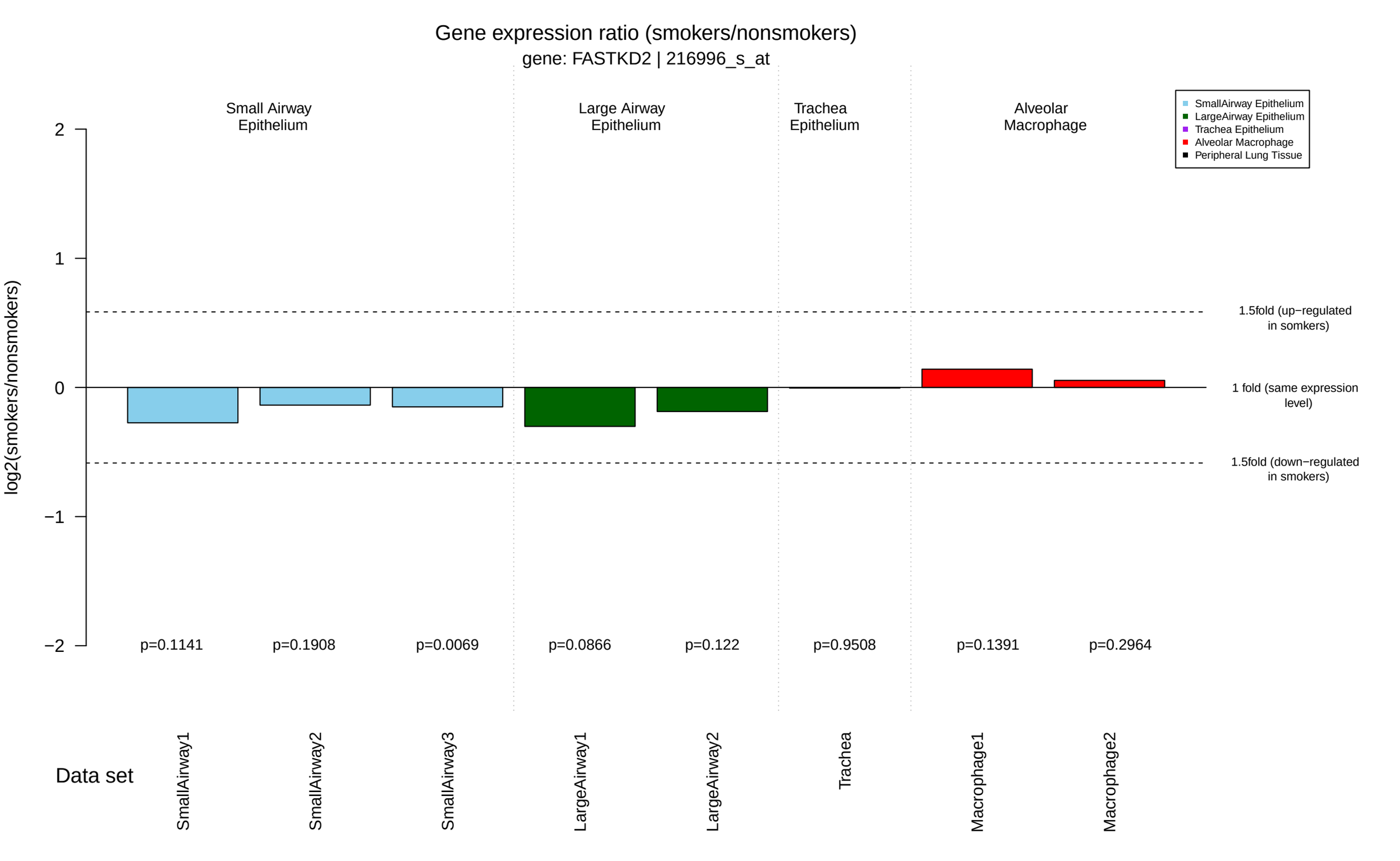
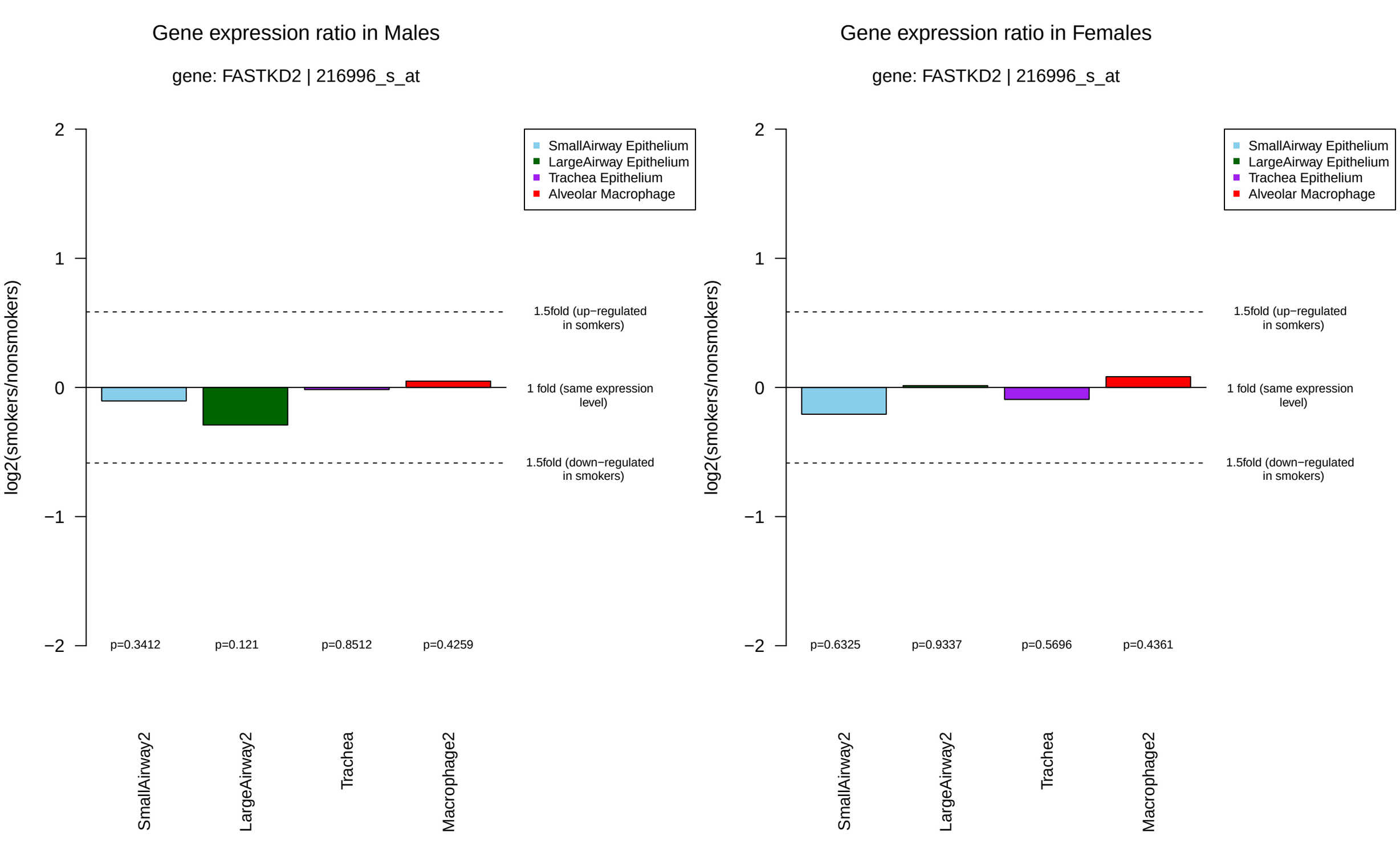
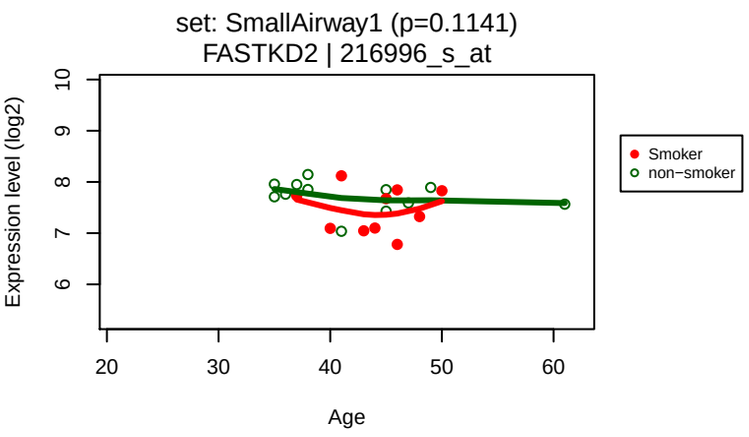
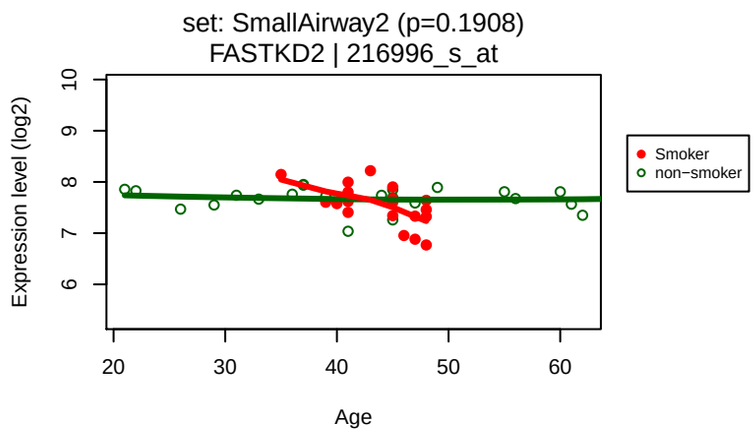
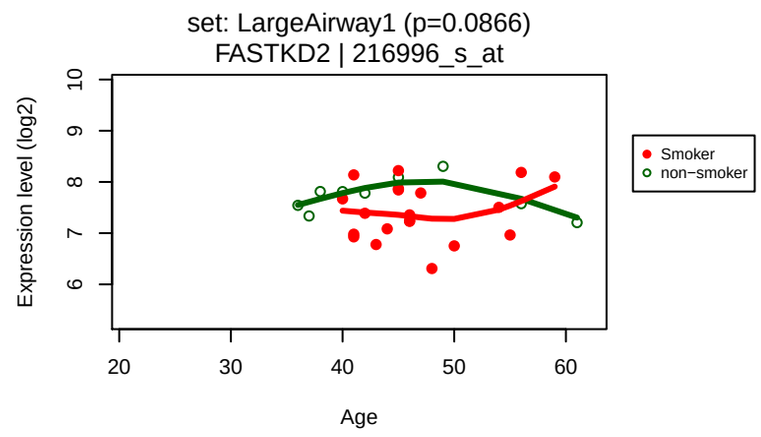

Gene expression of smokers and nonsmokers
gene: FASTKD2 | 216996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1141
p=0.1908
p=0.0069
p=0.0866
p=0.122
p=0.9508
p=0.1391
p=0.2964
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

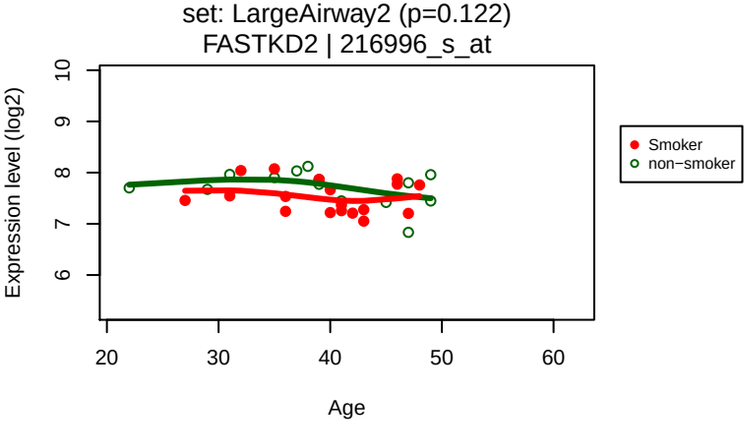

Gene expression ratio (smokers/nonsmokers)
gene: FASTKD2 | 216996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1141
p=0.1908
p=0.0069
p=0.0866
p=0.122
p=0.9508
p=0.1391
p=0.2964
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

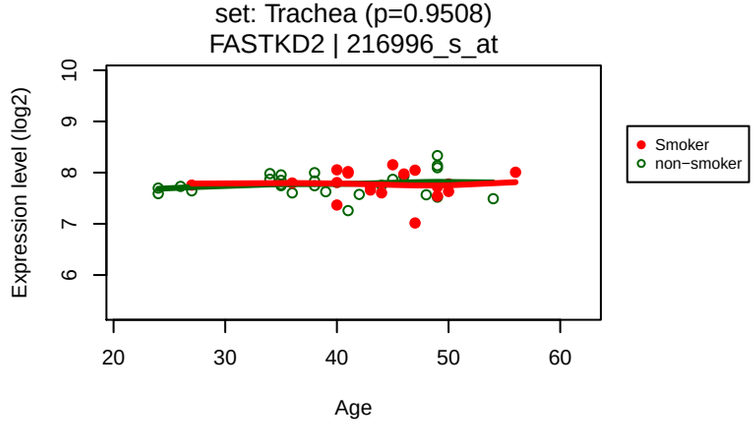

Gene expression ratio in Males
gene: FASTKD2 | 216996_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3412
p=0.121
p=0.8512
p=0.4259
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FASTKD2 | 216996_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6325
p=0.9337
p=0.5696
p=0.4361
SmallAirway2
LargeAirway2
Trachea
Macrophage2

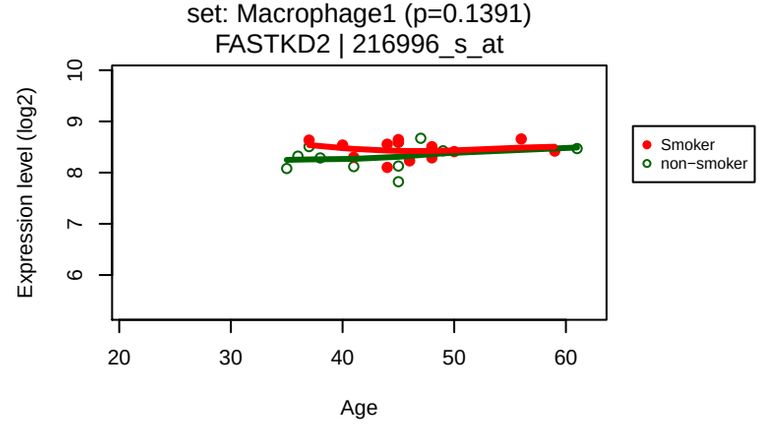

set: SmallAirway1 (p=0.1141)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker

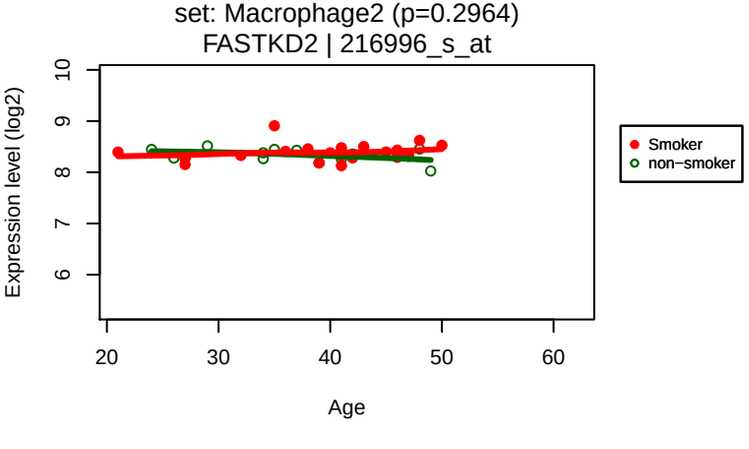

set: SmallAirway2 (p=0.1908)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker

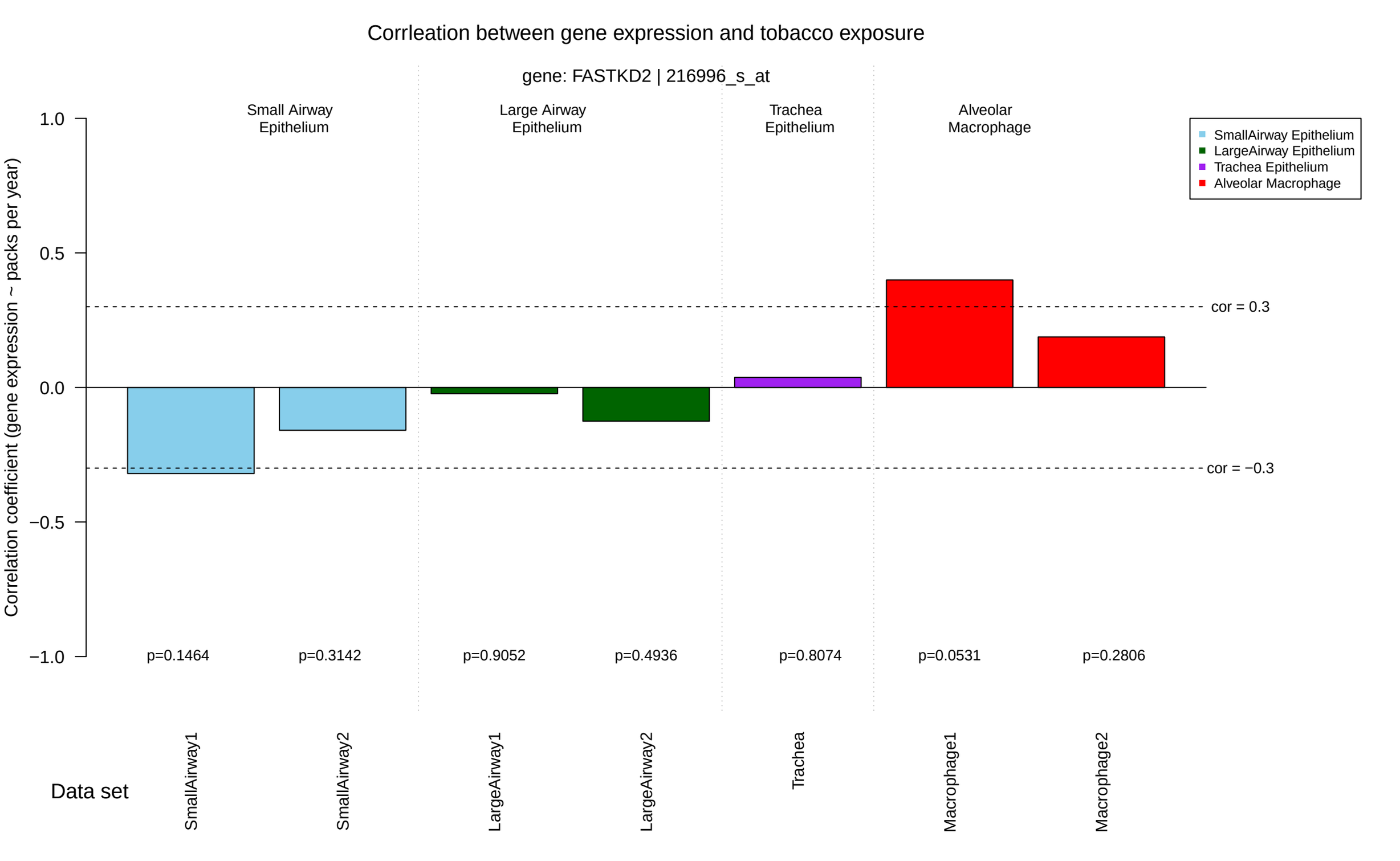

set: LargeAirway1 (p=0.0866)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.122)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9508)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1391)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2964)
FASTKD2 | 216996_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FASTKD2 | 216996_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1464
p=0.3142
p=0.9052
p=0.4936
p=0.8074
p=0.0531
p=0.2806
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2